Elementary Analysis

Section 27: Weierstrass's Approximation Theorem

Exercises 27.1–27.7, with every subpart
Detailed worked solutions

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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27 Weierstrass's approximation theorem

Approximation is different from equality with a power series. A continuous function on a closed bounded interval can be approximated uniformly by polynomials, even when it has corners. The approximating polynomials need not be successive partial sums of a single power series. The Bernstein polynomials on $[0,1]$ are

$$B_n f(x) = \sum_{k=0}^{n} f(k/n)\binom{n}{k}x^k(1-x)^{n-k} \qquad (n \geq 1).$$

For $x \in [0,1]$, the weights $\binom{n}{k}x^k(1-x)^{n-k}$ are nonnegative and sum to 1. Thus $B_n f(x)$ is a weighted average of sampled values of f. Products with exponent zero are interpreted as 1, including at endpoints.

27.1 Exercise 27.1: Transferring approximation from $[0,1]$ to $[a,b]$

What we must show. Deduce the general Weierstrass approximation theorem from the Bernstein theorem on $[0,1]$.

Move the input interval to the standard one.

Assume first $a < b$. The affine function $\phi(t) = a + (b-a)t$ maps $[0,1]$ onto $[a,b]$, with inverse $\phi^{-1}(x) = (x-a)/(b-a)$. If f is continuous on $[a,b]$, then $g = f \circ \phi$ is continuous on $[0,1]$. Theorem 27.4 supplies polynomials $q_n = B_n g$ converging uniformly to g there.

Move the approximating polynomials back.

Define

$$p_n(x) = q_n\left(\frac{x-a}{b-a}\right).$$

A polynomial composed with an affine function is again a polynomial. For every $x \in [a,b]$, let $t = (x-a)/(b-a) \in [0,1]$. Then

$$|p_n(x) - f(x)| = |q_n(t) - g(t)|.$$

Because the substitution is a bijection between the intervals, the suprema of these errors on the two intervals are equal. The supremum on $[0,1]$ tends to zero, so the one on $[a,b]$ does too.

Include the degenerate interval if needed.

If $a = b$, the constant polynomial $p_n(x) = f(a)$ is already exact on the one-point interval.

Conclusion. Every continuous function on $[a,b]$ is a uniform limit of polynomials.

27.2 Exercise 27.2: One sequence that works on every bounded subset of $\mathbb{R}$

What we must show. For continuous f on $\mathbb{R}$, find polynomials p_n converging uniformly to f on each bounded subset.

Use a larger interval and a smaller tolerance at each stage.

For each integer $n \geq 1$, Weierstrass approximation on $[-n, n]$ gives one polynomial p_n with

$$|p_n(x) - f(x)| < \frac{1}{n} \qquad (|x| \leq n).$$

We choose one such polynomial for each n.

Verify convergence on an arbitrary bounded set.

Let $E \subseteq \mathbb{R}$ be bounded. Choose M so that $|x| \leq M$ for every $x \in E$. Given $\varepsilon > 0$, choose an integer N larger than both M and $1/\varepsilon$. For $n > N$, every $x \in E$ belongs to $[-n, n]$, so $|p_n(x) - f(x)| < 1/n < \varepsilon$.

Conclusion. The constructed sequence converges uniformly on every bounded subset of $\mathbb{R}$. The index threshold can depend on that bounded set; this does not assert uniform convergence on all of $\mathbb{R}$.

27.3 Exercise 27.3: Why the entire real line is too large

What we must show. Show that neither $\sin x$ nor e^x is a uniform limit of polynomials on $\mathbb{R}$.

(a) A polynomial uniformly close to sine would have to be constant.

Suppose $p_n \to \sin$ uniformly. Then for some index m, $|p_m(x) - \sin x| < 1/2$ for every real x. Consequently $|p_m(x)| \leq |\sin x| + 1/2 \leq 3/2$ everywhere. A nonconstant polynomial is unbounded on $\mathbb{R}$: its leading term dominates all terms of smaller degree as $|x| \to \infty$. Thus p_m must be a constant c. At $x = \pi/2$ and $x = 3\pi/2$, the proposed approximation requires $|c - 1| < 1/2$ and $|c + 1| < 1/2$. But the triangle inequality would then give $2 \leq |1 - c| + |c + 1| < 1$, an impossibility.

(b) No polynomial can stay within a fixed error of the exponential.

Let p be any polynomial. Choose an integer $d \geq 0$ at least its degree, and $C > 0$ such that $|p(x)| \leq Cx^d$ for $x \geq 1$; the sum of the absolute values of its coefficients, enlarged if necessary, supplies such a C. The exponential series has nonnegative terms at nonnegative inputs, so

$$e^x \geq \frac{x^{d+1}}{(d+1)!} \qquad (x \geq 0).$$

Therefore, for $x \geq 1$,

$$e^x - p(x) \geq \frac{x^{d+1}}{(d+1)!} - Cx^d = x^d \left(\frac{x}{(d+1)!} - C \right) \longrightarrow +\infty.$$

In particular, no polynomial has $|p(x) - e^x| < 1$ for all real x. Uniform convergence would eventually require exactly such a polynomial, so it is impossible.

Conclusion. Uniform polynomial approximation on each bounded interval does not extend to uniform approximation on the whole real line in either example.

27.4 Exercise 27.4: Correcting the two endpoint values

What we must show. Construct approximating polynomials with $p_n(a) = f(a)$ and $p_n(b) = f(b)$.

Start with any uniformly approximating sequence.

For $a < b$, choose polynomials $q_n \to f$ uniformly on $[a, b]$. Write $E_n = \sup_{x\in[a,b]} |q_n(x) - f(x)|$, so $E_n \to 0$. The desired corrections at the two endpoints are $A_n = f(a) - q_n(a)$ and $B_n = f(b) - q_n(b)$.

Interpolate those corrections with a straight line.

Define

$$s_n(x) = \frac{b-x}{b-a}A_n + \frac{x-a}{b-a}B_n, \qquad p_n(x) = q_n(x) + s_n(x).$$

The function s_n is affine, so p_n is a polynomial. At $x = a$ its correction is A_n, and at $x = b$ its correction is B_n. Thus $p_n(a) = f(a)$ and $p_n(b) = f(b)$ exactly.

Show that the correction stays small throughout the interval.

On $[a, b]$ the two coefficients in s_n are nonnegative and sum to 1. Also $|A_n|, |B_n| \le E_n$. Hence $|s_n(x)| \le E_n$ for all such x, and

$$|p_n(x) - f(x)| \le |q_n(x) - f(x)| + |s_n(x)| \le 2E_n \longrightarrow 0.$$

The bound is uniform in x. If $a = b$, the constant polynomial $f(a)$ settles the one-point case directly.

Conclusion. The endpoint-corrected polynomials retain uniform convergence.

27.5 Exercise 27.5: Bernstein polynomials for the first two powers

What we must show. Compute $B_n f$ when $f(x) = x$ and when $f(x) = x^2$.

(a) Use the first binomial moment.

For $f(t) = t$,

$$B_n f(x) = \frac{1}{n}\sum_{k=0}^{n} k\binom{n}{k}x^k(1-x)^{n-k}.$$

For $k \ge 1$, $k\binom{n}{k} = n\binom{n-1}{k-1}$. Substitution and the binomial theorem give

$$\sum_{k=0}^{n} k\binom{n}{k}x^k(1-x)^{n-k} = nx\sum_{j=0}^{n-1}\binom{n-1}{j}x^j(1-x)^{n-1-j} = nx.$$

Therefore $B_n f(x) = x$ for every $n \ge 1$: linear functions of this form are reproduced exactly.

(b) Decompose the square into two binomial moments.

For $f(t) = t^2$, use $k^2 = k(k-1) + k$. For $n \ge 2$, the factorial identity $k(k-1)\binom{n}{k} = n(n-1)\binom{n-2}{k-2}$ gives

$$\sum_{k=0}^{n} k(k-1)\binom{n}{k}x^k(1-x)^{n-k} = n(n-1)x^2.$$

The same identity is zero on both sides for $n = 1$. Adding the first moment nx and dividing by n^2 yields

$$B_n f(x) = \frac{n(n-1)x^2 + nx}{n^2} = x^2 + \frac{x(1-x)}{n}.$$

As a check, $0 \le x(1-x) \le 1/4$ on $[0, 1]$, so the error is at most $1/(4n)$, consistent with uniform convergence.

Conclusion. $B_n(t \mapsto t)(x) = x$ and $B_n(t \mapsto t^2)(x) = x^2 + x(1-x)/n$.

27.6 Exercise 27.6: A necessary condition for uniform Bernstein approximation

What we must show. If $B_n f \to f$ uniformly on $[0,1]$, prove that f is continuous, without initially assuming continuity of f.

Notice what kind of function each approximant is.

For each fixed n, the finitely many sampled values $f(k/n)$ are real constants. Thus the formula defining $B_n f$ is a finite linear combination of polynomials, so $B_n f$ is a polynomial and is continuous on $[0,1]$.

Use the uniform-limit theorem.

The assumption is uniform convergence of this sequence of continuous functions to f. Theorem 24.3 therefore makes f continuous on $[0,1]$.

Conclusion. Continuity is necessary for uniform convergence of the Bernstein polynomials to the original function. Together with Theorem 27.4, it is also sufficient.

27.7 Exercise 27.7: A Bernstein bound from nonnegative weights

What we must show. If $|f(t)| \le M$ for every $t \in [0,1]$, prove $|B_n f(x)| \le M$ for all $n \ge 1$ and $x \in [0,1]$.

Use the triangle inequality with signs of the weights checked.

For $0 \le x \le 1$, every weight $w_k = \binom{n}{k} x^k (1-x)^{n-k}$ is nonnegative. Also each sample point k/n belongs to $[0,1]$, so $|f(k/n)| \le M$. Hence

$$|B_n f(x)| \le \sum_{k=0}^{n} |f(k/n)| w_k \le M \sum_{k=0}^{n} w_k.$$

Evaluate the total weight.

By the binomial theorem, $\sum_{k=0}^{n} w_k = (x + (1-x))^n = 1$. The preceding bound is therefore M.

Conclusion. Every Bernstein polynomial obeys the same bound M as f on $[0,1]$. The interval matters: outside it the weights need not be nonnegative.